CFTC Positioning Pattern Thresholds — Experiment Results for Sign-Off

1. Purpose

Grid-search results for **SQUEEZE** (FM low + RM high) and **LIQUIDITY EXIT** (RM low + FM high) before locking production Sentiment Layer 3 flags. **Display/alert only** — not SSI sizing gates (per 2 Aug email).

2. Data & methodology

Item	Value
CFTC Fast Money through	**2026-08-18** (Tuesday position date)
CFTC Real Money through	**2026-08-18**
Units	**emini_equivalent** — component contract lines at notional weight; CFTC's 2023-05-02 redefinition of the Consolidated line is no longer a seam
Raw weekly prints	2006-06-13 → 2026-08-18 (1054 weeks)
Percentile window	156 weeks, rolling — **partial windows are not ranked**
Backtest start	**2009-06-02** (first full window; 899 weeks)
Unit-break scan	none
Episode collapse	Consecutive qualifying weeks → one episode (first fire date)
Forward returns	S&P 500 at 4w / 8w / 12w trading days
Benchmark	Mean SPX return across **all** weeks in sample per horizon
Excess	Per-episode SPX return minus benchmark; excess_hit = beat market
Ranking metric	**Mean − median gap** (tail marker), not Sharpe
Per-cell columns	n_wk, n_ep, mean, median, gap, hit %, best, worst, top dated instances
SQUEEZE JSON	`03_squeeze_grid_v2_20260826.json`
LIQUIDITY EXIT JSON	`04_liquidity_exit_grid_v2_20260826.json`

SQUEEZE: FM pctile < X AND RM pctile > Y (same week).

LIQUIDITY EXIT: RM pctile < X AND FM pctile > Y.

Worked examples — why mean−median gap, not Sharpe or hit rate alone

Example A (tail-driven squeeze is real): +22%, +18%, +14%, +3%, +2%, +1%, −1%, −2%, −4%

- mean **5.8889%**, median **2.0%**, gap **3.8889%**, hit **66.67%**, worst **-4.0%**

Example B (nothing there — market beta): +6%, +5%, +4%, +3%, +3%, +2%, +1%, −1%, −2%

- mean **2.3333%**, median **3.0%**, gap **-0.6667%**, hit **77.78%**, worst **-2.0%**

B has higher hit rate and better worst case. Only the gap and dated top instances reveal A's tail edge.

PAR (unconditional — every week in sample, no episode collapse)

Excess = SPX forward return minus mean SPX return across **all** weeks at that horizon. excess_hit = share of observations that **beat the market** (excess > 0), not merely positive SPX.

- **4w** (bench 1.1040369950327467%): n_wk=896 | mean=1.104% med=1.5702% | mean_excess=0.0% med_excess=0.4662% | hit=68.97% excess_hit=56.7%
- **8w** (bench 2.1151226987168106%): n_wk=892 | mean=2.1151% med=2.6147% | mean_excess=0.0% med_excess=0.4995% | hit=72.76% excess_hit=54.15%
- **12w** (bench 3.1104724182402177%): n_wk=887 | mean=3.1105% med=3.8267% | mean_excess=-0.0% med_excess=0.7162% | hit=75.65% excess_hit=56.82%

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3. Executive summary

PAR (unconditional, 899 weeks): 12w excess-hit 56.82%. Every cell below is to be read against that number, not against its own win rate.

1. **The previously recommended cell, `FM<10 / RM>55`, is at par.** n_ep=7, mean=1.9266%, mean_excess=-1.1838%, excess_hit=57.14% against PAR 56.82%. It is not recommended.
2. **Highest excess-hit cell: `FM_roll_pct<5 AND RM_roll_pct>40`** — n_ep=5, mean=5.0201%, mean_excess=1.9096%, excess_hit=100.0%. Pre/post 2023-05-02: 3 vs 2 episodes, survives=True. **It does not survive out-of-sample testing** — see §3b: a random set of five episodes beats the market every time in 1 of 20 draws, the result inverts on a 104-week window, and the cell never fires in the first half of the sample at all.
3. **LIQUIDITY EXIT cannot fire, and it is the RM leg — not the FM units.** `RM_pct<30` covered 49.7% of weeks before 2023-05-02 and **0.0%** after; it last fired 2023-03-28, 177 weeks ago. `FM_pct>70` still fires (12.7% of post-2023 weeks). Asset managers have been persistently net long since 2023, so a rolling rank of RM no longer reaches its floor. Restating the units does not bring the pattern back.
4. **FM percentile has no linear relationship with forward SPX returns** at any horizon ($R^2 \leq 0.003$, $p \geq 0.10$). A threshold effect would not show in a linear fit, so this does not close the question — but it does settle row 42: `invert=True` on `cot_fast_money` is immaterial either way.

3b. Out-of-sample checks on the only above-par cells (FM<5)

Small n on its own is not a reason to dismiss these — the 4 Aug spec asked for exactly that shape. Four checks were run instead, and the cells fail three of them.

- **Placebo:** a *random* set of 4 episodes beats the market every time in **10.8%** of draws — 1 in 9. With 66 grid cells searched, the best one showing 100% is close to what chance produces.
- **Placebo:** a *random* set of 5 episodes beats the market every time in **5.1%** of draws — 1 in 20. With 66 grid cells searched, the best one showing 100% is close to what chance produces.
- **Window sensitivity:** on a 104-week window the same cell gives n=11, mean excess **-0.4464%**, hit 60.0% — the result inverts. A threshold effect should not depend on the window used to define the threshold.
- **Walk-forward:** split at 2018-01-09, the cell fires **zero times** in the first half and all of its episodes in the second. It has never been tested out of sample.
- **Neighbour stability:** FM<2.5 gives 66.67% and FM<6 gives 85.71%. 100% exists at exactly 5 and nowhere adjacent — the same objection raised against FM<10, and it applies here too.
- **On the bootstrap:** its interval for this cell excludes zero, which looks supportive. It is not evidence. The bootstrap resamples the cell's own episodes, so it measures sampling variability around a result it assumes is real; the placebo test is the one that asks whether the selection was lucky.

Full tables: `_generated/cftc_oos_fm5_check_20260826.md`

Recommendation

None — sign-off should stay held. After out-of-sample testing there is no candidate left. The FM<5 cells were the only ones above par and they fail the placebo, window-sensitivity and walk-forward checks (§3b). LIQUIDITY EXIT's trigger leg has been unreachable for three years and needs a spec decision, not a threshold. Shipping a placeholder in the meantime would put a flag on the page that the data does not support.

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4. SQUEEZE — full distribution per cell (12w, ranked by mean−median gap)

FM <	RM >	n_wk	n_ep	mean %	median %	gap %	hit %	mean_ex %	med_ex %	ex_hit %	best_for_signal %	worst %
5	45	11	5	5.3648	4.8651	0.4998	100.0	2.2544	1.7546	100.0	8.4371	3.292
5	50	10	4	6.0557	5.669	0.3867	100.0	2.9453	2.5585	100.0	8.4371	4.061

FM <	RM >	n_wk	n_ep	mean %	median %	gap %	hit %	mean_ex %	med_ex %	ex_hit %	best_for_signal %	worst %
5	40	12	5	5.0201	4.8651	0.155	100.0	1.9096	1.7546	100.0	7.058	3.292
12.5	50	26	11	3.5686	3.6058	-0.0372	90.91	0.4582	0.4953	72.73	11.2305	-14.19
30	45	120	31	3.1457	3.2921	-0.1464	80.65	0.0352	0.1816	54.84	12.3439	-15.02
30	40	122	31	3.1012	3.2921	-0.1909	80.65	-0.0093	0.1816	54.84	12.3439	-15.02
30	60	93	29	2.7562	3.0904	-0.3342	79.31	-0.3542	-0.0201	48.28	12.3439	-15.02
30	65	82	27	2.6501	3.0647	-0.4146	77.78	-0.4604	-0.0458	48.15	12.3439	-15.02
10	60	14	7	1.4696	2.1428	-0.6732	85.71	-1.6409	-0.9677	42.86	8.7829	-14.19
12.5	45	32	12	2.4182	3.2199	-0.8017	83.33	-0.6922	0.1094	58.33	8.7829	-14.19

FM <	RM >	n_wk	n_ep	mean %	median %	gap %	hit %	mean_ex %	med_ex %	ex_hit %	best_for_signal %	worst %
30	50	111	30	3.3384	4.1483	-0.8099	80.0	0.2279	1.0379	60.0	12.3439	-15.02
25	45	96	30	2.945	3.8183	-0.8733	83.33	-0.1654	0.7079	56.67	10.7851	-15.02
12.5	40	33	12	2.3033	3.2199	-0.9166	83.33	-0.8072	0.1094	58.33	8.7829	-14.19
25	60	71	25	2.3986	3.3172	-0.9186	80.0	-0.7118	0.2067	52.0	10.7851	-15.02
25	40	98	30	2.8991	3.8183	-0.9193	83.33	-0.2114	0.7079	56.67	10.7851	-15.02
12.5	65	16	8	1.7212	2.6452	-0.924	87.5	-1.3892	-0.4652	50.0	8.7829	-14.19
25	65	62	25	2.3165	3.3172	-1.0007	80.0	-0.794	0.2067	52.0	10.7851	-15.02
10	50	21	9	3.2241	4.2369	-1.0128	88.89	0.1136	1.1264	66.67	8.7829	-14.19
10	45	25	9	3.1892	4.2369	-1.0477	88.89	0.0787	1.1264	66.67	8.7829	-14.19

FM <	RM >	n_wk	n_ep	mean %	median %	gap %	hit %	mean_ex %	med_ex %	ex_hit %	best_for_signal %	worst %
10	65	13	6	1.0084	2.084	-1.0756	83.33	-2.1021	-1.0265	33.33	8.7829	-14.19
12.5	55	19	9	2.3562	3.4334	-1.0772	88.89	-0.7542	0.3229	66.67	8.7829	-14.19
20	50	62	24	2.2896	3.3753	-1.0857	87.5	-0.8208	0.2648	54.17	11.2305	-15.69
45	60	159	31	2.2838	3.4138	-1.13	80.65	-0.8267	0.3033	54.84	9.6364	-15.37
12.5	60	18	9	2.0008	3.1477	-1.1469	88.89	-1.1097	0.0372	55.56	8.7829	-14.19
30	55	102	30	2.9713	4.1483	-1.177	80.0	-0.1392	1.0379	56.67	12.3439	-15.02
25	50	87	29	3.1375	4.3195	-1.182	82.76	0.027	1.209	62.07	11.2305	-15.02
10	40	26	9	3.036	4.2369	-1.2009	88.89	-0.0745	1.1264	66.67	8.7829	-14.19

FM <	RM >	n_wk	n_ep	mean %	median %	gap %	hit %	mean_ex %	med_ex %	ex_hit %	best_for_signal %	worst %
20	45	69	24	1.8687	3.0739	-1.2053	87.5	-1.2418	-0.0365	50.0	8.4371	-15.69
20	40	71	24	1.8112	3.0739	-1.2627	87.5	-1.2993	-0.0365	50.0	8.0602	-15.69
40	60	138	30	2.7017	4.0764	-1.3747	86.67	-0.4088	0.9659	60.0	9.6364	-30.58
45	40	215	33	2.9914	4.3971	-1.4057	81.82	-0.1191	1.2866	66.67	12.8913	-15.37
20	55	55	22	1.6774	3.0865	-1.4091	86.36	-1.4331	-0.024	50.0	8.0602	-15.69
25	55	79	28	2.6714	4.1095	-1.4382	82.14	-0.4391	0.9991	57.14	10.7851	-15.02
45	45	209	33	2.8939	4.3755	-1.4816	81.82	-0.2166	1.265	63.64	12.8913	-15.37
40	65	123	27	2.9076	4.4764	-1.5688	88.89	-0.2028	1.3659	66.67	9.6364	-30.58

FM <	RM >	n_wk	n_ep	mean %	median %	gap %	hit %	mean_ex %	med_ex %	ex_hit %	best_for_signal %	worst %
45	65	141	30	2.6856	4.2924	-1.6069	83.33	-0.4249	1.182	63.33	9.6364	-15.37
45	50	193	30	3.1351	4.7694	-1.6343	80.0	0.0246	1.6589	70.0	12.8913	-15.37
7.5	50	18	7	2.5844	4.2369	-1.6525	85.71	-0.526	1.1264	71.43	8.7119	-14.19
35	45	145	32	1.9229	3.6178	-1.6948	81.25	-1.1875	0.5073	59.38	9.6364	-30.58
7.5	45	20	7	2.5396	4.2369	-1.6973	85.71	-0.5709	1.1264	71.43	8.7119	-14.19
35	40	147	32	1.8798	3.6178	-1.7379	81.25	-1.2306	0.5073	59.38	9.6364	-30.58
35	60	113	30	1.4512	3.1904	-1.7391	80.0	-1.6592	0.0799	53.33	9.6364	-30.58
20	65	43	20	1.5815	3.3753	-1.7938	85.0	-1.529	0.2648	55.0	8.7829	-15.69
20	60	50	19	1.4719	3.3172	-1.8453	84.21	-1.6386	0.2067	52.63	8.0602	-15.69

FM <	RM >	n_wk	n_ep	mean %	median %	gap %	hit %	mean_ex %	med_ex %	ex_hit %	best_for_signal %	worst %
40	50	164	28	3.4423	5.2976	-1.8552	85.71	0.3318	2.1871	75.0	11.7032	-30.58
7.5	40	21	7	2.3426	4.2369	-1.8943	85.71	-0.7679	1.1264	71.43	8.7119	-14.19
40	45	176	30	3.1768	5.1493	-1.9725	86.67	0.0663	2.0388	66.67	11.7032	-30.58
40	40	181	29	3.1539	5.1569	-2.003	86.21	0.0434	2.0464	68.97	11.7032	-30.58
15	50	39	19	1.4203	3.4334	-2.0131	73.68	-1.6902	0.3229	57.89	11.2305	-15.69
35	50	135	30	2.0872	4.1703	-2.0831	80.0	-1.0233	1.0598	66.67	11.2305	-30.58
45	55	179	30	2.7709	4.8933	-2.1224	80.0	-0.3396	1.7828	66.67	11.7032	-15.37
7.5	60	11	5	-0.1277	2.0252	-2.1529	80.0	-3.2382	-1.0853	40.0	5.669	-14.19

FM <	RM >	n_wk	n_ep	mean %	median %	gap %	hit %	mean_ex %	med_ex %	ex_hit %	best_for_signal %	worst %
40	55	153	28	2.9371	5.1493	-2.2122	85.71	-0.1734	2.0388	71.43	11.7032	-30.58
15	45	46	19	0.8885	3.1477	-2.2592	73.68	-2.2219	0.0372	52.63	8.7829	-15.69
10	55	15	7	1.9266	4.2369	-2.3103	85.71	-1.1838	1.1264	57.14	8.7829	-14.19
15	40	47	19	0.8159	3.1477	-2.3318	73.68	-2.2945	0.0372	52.63	8.7829	-15.69
35	65	100	27	1.2061	3.6419	-2.4358	77.78	-1.9044	0.5314	55.56	9.6364	-30.58
35	55	125	31	1.8145	4.3971	-2.5826	80.65	-1.296	1.2866	64.52	9.6364	-30.58
15	55	32	17	0.5257	3.1477	-2.622	70.59	-2.5848	0.0372	52.94	8.7829	-15.69
15	65	26	16	-0.0925	2.5865	-2.679	68.75	-3.203	-0.524	50.0	8.7829	-15.69

FM <	RM >	n_wk	n_ep	mean %	median %	gap %	hit %	mean_ex %	med_ex %	ex_hit %	best_for_signal %	worst %
15	60	31	17	0.3375	3.0386	-2.7011	70.59	-2.773	-0.0719	47.06	8.7829	-15.69
7.5	65	10	4	-1.2189	1.8241	-3.043	75.0	-4.3293	-1.2863	25.0	5.669	-14.19
7.5	55	12	5	0.5121	4.2369	-3.7248	80.0	-2.5983	1.1264	60.0	5.669	-14.19

SQUEEZE heatmap — 12w mean−median gap

FM cut	RM>40	RM>45	RM>50	RM>55	RM>60	RM>65
FM<5	0.155	0.4998	0.3867	0.0	0.0	0.0
FM<7.5	-1.8943	-1.6973	-1.6525	-3.7248	-2.1529	-3.043
FM<10	-1.2009	-1.0477	-1.0128	-2.3103	-0.6732	-1.0756
FM<12.5	-0.9166	-0.8017	-0.0372	-1.0772	-1.1469	-0.924
FM<15	-2.3318	-2.2592	-2.0131	-2.622	-2.7011	-2.679
FM<20	-1.2627	-1.2053	-1.0857	-1.4091	-1.8453	-1.7938
FM<25	-0.9193	-0.8733	-1.182	-1.4382	-0.9186	-1.0007
FM<30	-0.1909	-0.1464	-0.8099	-1.177	-0.3342	-0.4146
FM<35	-1.7379	-1.6948	-2.0831	-2.5826	-1.7391	-2.4358
FM<40	-2.003	-1.9725	-1.8552	-2.2122	-1.3747	-1.5688
FM<45	-1.4057	-1.4816	-1.6343	-2.1224	-1.13	-1.6069

SQUEEZE heatmap — 12w excess_hit % (compare to PAR above)

FM cut	RM>40	RM>45	RM>50	RM>55	RM>60	RM>65
FM<5	100.0 (n=5)	100.0 (n=5)	100.0 (n=4)	100.0 (n=3)	100.0 (n=2)	100.0 (n=2)
FM<7.5	71.43 (n=7)	71.43 (n=7)	71.43 (n=7)	60.0 (n=5)	40.0 (n=5)	25.0 (n=4)
FM<10	66.67 (n=9)	66.67 (n=9)	66.67 (n=9)	57.14 (n=7)	42.86 (n=7)	33.33 (n=6)
FM<12.5	58.33 (n=12)	58.33 (n=12)	72.73 (n=11)	66.67 (n=9)	55.56 (n=9)	50.0 (n=8)

FM cut	RM>40	RM>45	RM>50	RM>55	RM>60	RM>65
FM<15	52.63 (n=19)	52.63 (n=19)	57.89 (n=19)	52.94 (n=17)	47.06 (n=17)	50.0 (n=16)
FM<20	50.0 (n=24)	50.0 (n=24)	54.17 (n=24)	50.0 (n=22)	52.63 (n=19)	55.0 (n=20)
FM<25	56.67 (n=30)	56.67 (n=30)	62.07 (n=29)	57.14 (n=28)	52.0 (n=25)	52.0 (n=25)
FM<30	54.84 (n=31)	54.84 (n=31)	60.0 (n=30)	56.67 (n=30)	48.28 (n=29)	48.15 (n=27)
FM<35	59.38 (n=32)	59.38 (n=32)	66.67 (n=30)	64.52 (n=31)	53.33 (n=30)	55.56 (n=27)
FM<40	68.97 (n=29)	66.67 (n=30)	75.0 (n=28)	71.43 (n=28)	60.0 (n=30)	66.67 (n=27)
FM<45	66.67 (n=33)	63.64 (n=33)	70.0 (n=30)	66.67 (n=30)	54.84 (n=31)	63.33 (n=30)

SQUEEZE heatmap — 12w mean % (n_ep in parentheses)

FM cut	RM>40	RM>45	RM>50	RM>55	RM>60	RM>65
FM<5	5.0201 (n=5)	5.3648 (n=5)	6.0557 (n=4)	5.4467 (n=3)	5.669 (n=2)	5.669 (n=2)
FM<7.5	2.3426 (n=7)	2.5396 (n=7)	2.5844 (n=7)	0.5121 (n=5)	-0.1277 (n=5)	-1.2189 (n=4)
FM<10	3.036 (n=9)	3.1892 (n=9)	3.2241 (n=9)	1.9266 (n=7)	1.4696 (n=7)	1.0084 (n=6)
FM<12.5	2.3033 (n=12)	2.4182 (n=12)	3.5686 (n=11)	2.3562 (n=9)	2.0008 (n=9)	1.7212 (n=8)
FM<15	0.8159 (n=19)	0.8885 (n=19)	1.4203 (n=19)	0.5257 (n=17)	0.3375 (n=17)	-0.0925 (n=16)
FM<20	1.8112 (n=24)	1.8687 (n=24)	2.2896 (n=24)	1.6774 (n=22)	1.4719 (n=19)	1.5815 (n=20)
FM<25	2.8991 (n=30)	2.945 (n=30)	3.1375 (n=29)	2.6714 (n=28)	2.3986 (n=25)	2.3165 (n=25)
FM<30	3.1012 (n=31)	3.1457 (n=31)	3.3384 (n=30)	2.9713 (n=30)	2.7562 (n=29)	2.6501 (n=27)
FM<35	1.8798 (n=32)	1.9229 (n=32)	2.0872 (n=30)	1.8145 (n=31)	1.4512 (n=30)	1.2061 (n=27)
FM<40	3.1539 (n=29)	3.1768 (n=30)	3.4423 (n=28)	2.9371 (n=28)	2.7017 (n=30)	2.9076 (n=27)
FM<45	2.9914 (n=33)	2.8939 (n=33)	3.1351 (n=30)	2.7709 (n=30)	2.2838 (n=31)	2.6856 (n=30)

Episode count (n_ep / n_wk)

FM cut	RM>40	RM>45	RM>50	RM>55	RM>60	RM>65
FM<5	5/12	5/11	4/10	3/8	2/7	2/7
FM<7.5	7/21	7/20	7/18	5/12	5/11	4/10
FM<10	9/26	9/25	9/21	7/15	7/14	6/13
FM<12.5	12/33	12/32	11/26	9/19	9/18	8/16
FM<15	19/47	19/46	19/39	17/32	17/31	16/26
FM<20	24/71	24/69	24/62	22/55	19/50	20/43
FM<25	30/98	30/96	29/87	28/79	25/71	25/62
FM<30	31/122	31/120	30/111	30/102	29/93	27/82
FM<35	32/147	32/145	30/135	31/125	30/113	27/100
FM<40	29/181	30/176	28/164	28/153	30/138	27/123

FM cut	RM>40	RM>45	RM>50	RM>55	RM>60	RM>65
FM<45	33/215	33/209	30/193	30/179	31/159	30/141

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5. LIQUIDITY EXIT — top cells by 4w mean−median gap (RM ≤ 35, FM ≥ 45)

`best_for_signal` / `worst_for_signal` are relative to the side the pattern trades. SQUEEZE is long, so best = highest forward return. LIQUIDITY EXIT is short, so best = *most negative* forward return. The two tables therefore order these columns in opposite directions on purpose.

RM <	FM >	n_wk	n_ep	mean %	median %	gap %	hit %	mean_ex %	med_ex %	ex_hit %	best_for_signal %	worst_ %
15	75	39	15	0.1337	1.4562	-1.3225	40.0	-0.9704	0.3522	46.67	-10.5122	10.049
35	45	219	48	0.7447	1.9759	-1.2312	35.42	-0.3593	0.8718	39.58	-10.7794	8.8998
30	45	198	48	0.1286	1.3	-1.1714	39.58	-0.9754	0.196	47.92	-10.7794	7.3993
35	70	116	33	0.4116	1.5739	-1.1623	33.33	-0.6924	0.4699	45.45	-10.5122	6.1655
25	45	178	49	0.2077	1.3512	-1.1435	38.78	-0.8964	0.2472	46.94	-10.7794	7.3993
30	70	102	32	-0.0077	1.1178	-1.1255	34.38	-1.1117	0.0138	50.0	-10.5122	6.1655
15	45	101	30	0.3175	1.375	-1.0576	43.33	-0.7866	0.271	46.67	-10.7794	10.049
35	60	155	42	1.069	2.1118	-1.0428	26.19	-0.035	1.0078	38.1	-10.7419	8.6265
35	55	174	46	0.5891	1.6254	-1.0363	34.78	-0.515	0.5214	43.48	-10.7794	8.6265
25	70	90	32	0.1045	1.1363	-1.0318	34.38	-0.9995	0.0323	50.0	-10.5122	6.4949
35	75	90	31	1.3582	2.3658	-1.0076	35.48	0.2542	1.2618	38.71	-10.5122	10.049
15	50	92	27	0.801	1.795	-0.994	37.04	-0.3031	0.691	40.74	-10.7794	10.049
35	50	197	51	0.6915	1.6769	-0.9854	37.25	-0.4126	0.5729	45.1	-10.7794	8.6265
15	55	77	25	0.6995	1.6769	-0.9774	32.0	-0.4046	0.5729	40.0	-10.7794	10.049
35	65	132	40	1.1858	2.1118	-0.926	25.0	0.0818	1.0078	37.5	-10.5122	8.6265

Dated instances for the cells above (most favourable first for a short)

- `RM_roll_pct<15 AND FM_roll_pct>75`: 2022-04-12 -10.5122%; 2015-12-08 -5.8393%; 2022-04-26 -5.5978%; 2022-02-08 -5.3889%; 2016-01-12 -4.4783%; 2016-04-26 -0.7477%; 2014-12-16 +1.0103%; 2015-04-21 +1.4562%; 2018-10-30 +2.2799%
- `RM_roll_pct<35 AND FM_roll_pct>45`: 2022-04-12 -10.5122%; 2011-07-05 -6.2659%; 2012-04-17 -4.3228%; 2011-05-03 -3.1011%; 2012-03-13 -1.9514%; 2015-11-24 -1.1895%; 2012-04-03 -0.7832%; 2018-07-31 +2.8843%; 2021-11-30 +4.9499%
- `RM_roll_pct<30 AND FM_roll_pct>45`: 2022-04-12 -10.5122%; 2011-07-05 -6.2659%; 2012-04-17 -4.3228%; 2018-10-23 -3.6049%; 2011-05-03 -3.1011%; 2012-03-13 -1.9514%; 2015-11-24 -1.1895%; 2012-04-03 -0.7832%; 2021-12-07 +0.2951%

- `RM_roll_pct<35 AND FM_roll_pct>70`: 2022-04-12 -10.5122%; 2012-05-01 -6.5798%; 2015-12-08 -5.8393%; 2018-10-23 -3.6049%; 2015-06-02 -2.2037%; 2015-04-14 +0.1565%; 2015-05-12 +0.2896%; 2018-07-31 +2.8843%; 2021-11-30 +4.9499%
- `RM_roll_pct<25 AND FM_roll_pct>45`: 2015-07-28 -10.7794%; 2022-04-12 -10.5122%; 2011-07-05 -6.2659%; 2012-04-17 -4.3228%; 2018-10-23 -3.6049%; 2012-03-13 -1.9514%; 2015-11-24 -1.1895%; 2012-04-03 -0.7832%; 2021-12-14 +1.9909%
- `RM_roll_pct<30 AND FM_roll_pct>70`: 2022-04-12 -10.5122%; 2012-05-01 -6.5798%; 2015-12-08 -5.8393%; 2018-10-23 -3.6049%; 2015-06-02 -2.2037%; 2015-04-14 +0.1565%; 2015-05-12 +0.2896%; 2021-12-07 +0.2951%; 2014-11-04 +3.0928%
- `RM_roll_pct<15 AND FM_roll_pct>45`: 2015-07-28 -10.7794%; 2022-04-12 -10.5122%; 2010-05-11 -8.6608%; 2011-07-05 -6.2659%; 2022-02-08 -5.3889%; 2015-08-18 -4.8457%; 2015-06-02 -2.2037%; 2015-11-24 -1.1895%; 2018-10-30 +2.2799%
- `RM_roll_pct<35 AND FM_roll_pct>60`: 2011-07-12 -10.7419%; 2022-04-12 -10.5122%; 2012-05-01 -6.5798%; 2018-10-23 -3.6049%; 2015-12-01 -1.8677%; 2012-04-03 -0.7832%; 2018-07-31 +2.8843%; 2012-02-28 +2.9398%; 2021-11-30 +4.9499%
- `RM_roll_pct<35 AND FM_roll_pct>55`: 2022-04-12 -10.5122%; 2011-07-05 -6.2659%; 2012-04-17 -4.3228%; 2018-10-23 -3.6049%; 2011-05-03 -3.1011%; 2015-12-01 -1.8677%; 2012-04-03 -0.7832%; 2018-07-31 +2.8843%; 2021-11-30 +4.9499%
- `RM_roll_pct<25 AND FM_roll_pct>70`: 2022-04-12 -10.5122%; 2012-05-01 -6.5798%; 2015-12-08 -5.8393%; 2018-10-23 -3.6049%; 2015-06-02 -2.2037%; 2015-04-14 +0.1565%; 2015-05-12 +0.2896%; 2021-12-14 +1.9909%; 2014-11-04 +3.0928%
- `RM_roll_pct<35 AND FM_roll_pct>75`: 2022-04-12 -10.5122%; 2015-12-08 -5.8393%; 2022-04-26 -5.5978%; 2018-10-23 -3.6049%; 2022-01-25 -3.0059%; 2015-05-19 -1.2872%; 2018-07-31 +2.8843%; 2014-11-04 +3.0928%; 2021-11-30 +4.9499%
- `RM_roll_pct<15 AND FM_roll_pct>50`: 2015-07-28 -10.7794%; 2022-04-12 -10.5122%; 2011-07-05 -6.2659%; 2022-02-08 -5.3889%; 2015-08-18 -4.8457%; 2015-06-02 -2.2037%; 2015-11-24 -1.1895%; 2010-05-25 +1.6769%; 2018-10-30 +2.2799%
- `RM_roll_pct<35 AND FM_roll_pct>50`: 2022-04-12 -10.5122%; 2011-07-05 -6.2659%; 2012-04-17 -4.3228%; 2011-05-03 -3.1011%; 2012-03-13 -1.9514%; 2015-11-24 -1.1895%; 2012-04-03 -0.7832%; 2018-07-31 +2.8843%; 2021-11-30 +4.9499%
- `RM_roll_pct<15 AND FM_roll_pct>55`: 2015-07-28 -10.7794%; 2022-04-12 -10.5122%; 2011-07-05 -6.2659%; 2022-02-08 -5.3889%; 2015-06-02 -2.2037%; 2015-03-03 -1.8925%; 2015-12-01 -1.8677%; 2010-05-25 +1.6769%; 2018-10-30 +2.2799%
- `RM_roll_pct<35 AND FM_roll_pct>65`: 2022-04-12 -10.5122%; 2012-05-01 -6.5798%; 2015-12-08 -5.8393%; 2018-10-23 -3.6049%; 2015-06-02 -2.2037%; 2015-04-14 +0.1565%; 2015-05-12 +0.2896%; 2018-07-31 +2.8843%; 2021-11-30 +4.9499%

PDF default RM<30, FM>60

- **4w:** n_ep=44, mean=0.6331%, median=1.4554%, gap=-0.8223%, hit=29.55%, best=-10.7419%, worst=8.6265%
- **8w:** n_ep=44, mean=1.4141%, median=2.3492%, gap=-0.9351%, hit=34.09%, best=-9.9708%, worst=8.9465%
- **12w:** n_ep=44, mean=3.0149%, median=4.3232%, gap=-1.3083%, hit=25.0%, best=-12.9115%, worst=12.7985%

- top 4w: 2012-02-28 +2.9398%; 2016-07-19 +0.6641%; 2021-12-07 +0.2951%; 2012-04-03 -0.7832%; 2015-12-01 -1.8677%; 2018-10-23 -3.6049%; 2012-05-01 -6.5798%; 2022-04-12 -10.5122%; 2011-07-12 -10.7419%

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6. Sign-off

SQUEEZE: FM < ___ , RM > ___ (rank by mean−median gap + dated instances)

LIQUIDITY EXIT: RM < ___ , FM > ___

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7. Reproduce

```
cd MindWealth_UI
.venv/bin/python scripts/run_cftc_rohit_rerun.py
.venv/bin/python scripts/compile_cftc_pattern_threshold_report.py
```